SPECIFICATION

Background of the Invention

In conventional networking, the physical layer is fixed and binary; it transmits 1s and 0s. The interpretation of these bits into characters or data structures is handled at higher levels of the OSI model, most notably the Application Layer (Layer 7). For example, the HTTP protocol uses headers to declare the character encoding of the content (e.g., 'Content-Type: text/html; charset=UTF-8'). This approach is sufficient when the physical layer is dumb, but it is inadequate for new high-cardinality physical layers, where the "language" of the physical signals can change. There is no existing protocol to allow hardware to negotiate the meaning of its own physical states before data transmission begins.

Summary of the Invention

The present invention is a protocol, or handshake process, that allows two connected devices to dynamically negotiate and agree upon a data encoding map at the Physical and Data-Link layers (Layers 1 and 2). This enables the use of high-cardinality interfaces where the meaning of a physical state is not fixed.

The protocol operates as follows:

1. **Initialization:** When two devices are physically connected, they begin communication using a pre-defined, "default" encoding map. This default map is a low-density, universally supported standard (e.g., mapping a single bit to a black or white pixel state) to guarantee the initial handshake can be understood.

2. **Capabilities Advertisement:** The initiating device sends a "capabilities frame" across the link using this default encoding. This frame contains a list of all the high-density encoding maps it supports (e.g., "Standard Machine Code 24-bit," "UTF-8 Retrofit," "Session-Specific Disposable Map XYZ").

3. **Selection and Acknowledgment:** The receiving device reads the capabilities frame and selects a compatible encoding map from the list that it also supports. It sends a "selection frame" back to the initiating device, indicating its choice.

4. **Protocol Switch:** Once the selection is acknowledged, both devices load the selected high-density encoding map and switch to using it for the remainder of the communication session. All subsequent frames are interpreted according to this negotiated map.

This lower-layer negotiation is a fundamental requirement for unlocking the potential of variable-encoding hardware.

Brief Description of the Drawings

FIG. 1 is a sequence diagram illustrating the steps of the lower-layer encoding negotiation protocol.

Detailed Description of the Invention

Referring to **FIG. 1**, a sequence diagram shows the negotiation process between two devices, Device A (101) and Device B (102), connected by a high-cardinality parallel link. First, a physical connection is established (103). Immediately following, Device A sends a capabilities frame (104) to Device B. This frame is encoded using a universal default, low-density map

and lists the high-density maps supported by Device A. Device B receives this frame and compares the list to its own supported maps. Device B then sends a selection frame (105) back to Device A, specifying the chosen encoding map for the session. Upon receiving the selection, Device A sends an acknowledgment (ACK) frame (106). Once the handshake is complete, both devices switch their physical layer encoders and decoders to use the selected map, and the main data transmission begins (107) using the efficient, high-density encoding.